

Second assignment

The main **goal** of the second assignment is to search for information about design decisions in issue tracking systems. Specifically, you will use four approaches:

- a) Search engine.
- b) Knowledge graph.
- c) Code clustering (based on your results from the 1st assignment).
- d) Vanilla LLM.

The assignment will be done individually. Thus, during the assignment, you will try to find design decisions in twelve tasks. Each task is associated with an approach. Even though all students will solve all 12 tasks, each student will use different approaches for different tasks, and in different sequences.

Very important: please check from the tasks assignments sheet, which approaches you need to use for which task, and in which sequence you need to solve the tasks.

For example, if your sequence is provided as below, then you will solve tasks of Tika 1 using a search engine, but tasks of Tika 2 using a knowledge graph, while tasks of Lucene using an LLM...

Tika 1	Tika 2	JClouds	Mapreduce	Lucene	Yarn
Search engine	Knowledge graph	Clustered code	Search engine	Vanilla LLM	Search engine

To solve the tasks for the specified projects, open the respective document of each project, you will find two tasks per project. Each task has three questions.

Because each approach is different, we created separate instructions to use each approach. Thus, before using any of these approaches please carefully read its instructions.

Search engine instructions

- a) Open the URL of the system.

(Please note that: Your browser may display a security warning when accessing the website. If this occurs, please proceed to the site, as it is safe to use.)

- b) Select the project name relevant to the task you will solve. Login to the system using the User ID provided to you in the tasks assignments sheet. The User ID and the password are the same. You will be presented with the list of tasks assigned to you. Please first check that the tasks align with the tasks assigned to you. If not, please

let us know. Please solve the tasks in their provided sequence in the [tasks assignments sheet](#). Please make sure you use your own User ID and not others.

- b) Each task has a set of questions. First, read each question carefully. Some search approaches require keywords, others require questions.
- If keywords are required, then you need to write keywords to search. You need to submit at least 2 rated queries per question. For each query, evaluate the relevance of each returned issue based on the following Likert scale:
 - *Very relevant* (5): The issue contains sufficient information to answer the entire question.
 - *Relevant* (4): The issue contains information to answer part of the question.
 - *Distantly relevant* (3): The issue does not contain information to answer the question. But the content of the issue can help to refine the search.
 - *Less relevant* (2): The issue does not contain information to answer the question. But the content of the issue can be related to the question.
 - *Not relevant* (1): The issue has no relation to the question.
 - If the associated approach requires a question, then you need to write a question in the text box that helps you answer the question in a task. For each returned issue, please evaluate their relevance according to the Likert scale mentioned above. **Rate the results of at least 2 questions**.
 - After rating all results of a query, please download the ratings file using the button in the interface. Please note that you can still download the ratings by the end of the task. Downloading the rating is important as a proof of doing the task, which needs to be submitted in Panda.
 - After searching for answers to a question, provide an answer to each question in a task by filling the **Result Sheet** with the required fields. Furthermore, record evidence including (issue IDs, sentences or paragraphs from issues that helped you to answer a question). Please try to find (as much as possible) information from issues to answer the questions.

Important note: Try to memorize which projects were the easiest and hardest to answer their questions, as this will be asked in the *exit questionnaire*.

Knowledge graph instructions

- Please watch the **tutorial video** <https://youtu.be/pSw4UTpe-ho> first.
- You will be provided with a dedicated html-file containing the tasks and the information. This is a static website you must open locally on your desktop browser. Do not switch devices as the state is locally stored on your device. **Do not delete your browser data or cookies**. Be sure to complete all steps, your browser will download a file in the end, which you will have to submit in PANDA.

- Before you can start, you will be asked to fill in your User ID and rate your own experience in Software Architecture
- Each task has a scenario and a set of questions. Read each question carefully.
 - You will be asked to first rate the presented information and then find an answer to the question. Please do not use additional sources of information. Your answers will NOT be graded for correctness.
 - Before the first real task, you will receive a tutorial task to familiarize yourself with the tool.
- For each question you will be presented with several different 'topics':
 - Each topic represents extracted information from issues that are related to the question to a varying degree.
 - You will see three different tabs for each topic: Source Code Files, Attachments, and Entities. These represent different extracted information
 - Click on the nodes and edges to retrieve additional information.
 - Please consider the aforementioned tutorial video <https://youtu.be/pSw4UTpe-ho>.
- Go over each topic and consider the presented information.
 - On the topic tab, on the right side, give the information presented in the topic a relevancy rating according to the following likert scale:
 - *Very relevant* (5): The knowledge graph contains sufficient information to answer the entire question.
 - *Relevant* (4): The knowledge graph contains information to answer part of the question.
 - *Distantly relevant* (3): The knowledge graph does not contain information to answer the question, but the content relates vaguely to it.
 - *Less relevant* (2): The knowledge graph does not contain information to answer the question, but some entities relate to it.
 - *Not relevant* (1): The knowledge graph has no relation to the question.
 - Answer the question visible in the "Task" tab within the provided **Result Sheet**. If you are not able to answer it with the information presented to you, please state so. Please do not use additional sources of information. Your answers will NOT be graded for correctness.
 - Due to the randomization of the task order, verify that the task title matches the title within the provided **Result Sheet**.
- After each question answer for a Task has been submitted, you will be presented with some questionnaires. Please read the questions thoroughly and then click on the related fields.
 - In the NASA TLX questionnaire you will be asked to click on a linear scale to give a rating.
 - In the SUS questionnaire you will select a score for each question

- A new task will open. After all tasks have been completed you will be presented with a download button. Press it and store the downloaded file on your computer.
- Submit the file on PANDA, do not rename it.

Important note: Try to memorize if the project was easier or harder to answer its questions compared to other projects, as this will be asked in the *exit questionnaire*.

Clustered source code instructions

1. Open the provided HTML file for the assigned **project** and **algorithm**, shown in the tasks assignments sheet, in any web browser. Please first check that the tasks that appear align with the tasks assigned to you. If not, please let us know.
2. At the top of the page, you will see the project name, next to it, you will see the available task buttons. Use the task buttons to switch between the assigned project's tasks.
3. On the **Left Sidebar**, you will see:
 - A list of generated clusters: each cluster is identified by a cluster ID
 - A progress section that shows one progress row for each task: each row contains a series of dots, where each dot represents one cluster.
 - i. A **gray dot** means the cluster has not yet been fully evaluated for that task.
 - ii. A **green dot** means all relevancy questions for that cluster have been answered for that task.
4. The center workspace displays the selected cluster and the currently selected task. For each selected cluster, you will see:
 - The LLM-generated **Cluster Title**.
 - The LLM-generated **Cluster Description**.
 - A collapsible **Source Code files** section.
 - The current **Task Scenario**.
 - The three task questions:
 - i. **Question 1 (Existence):** Architectural components and connectors.
 - ii. **Question 2 (Property):** Solutions for quality attributes (patterns/tactics).
 - iii. **Question 3 (Executive):** Integrated tools and technologies.

You must evaluate **every cluster for every task** by rating the relevancy of the cluster's title and description to the three Task Questions.

5. For each Task Question:

- Rate the relevance of the current cluster to the question using the **Likert Scale** stated here below and also pinned in the **Right Sidebar** for your reference. Select one relevancy score from **1** to **5**. (The page automatically records your selected score when you click it.).
 - **Very relevant (5)**: The cluster title and description contains sufficient information to answer the entire question.
 - **Relevant (4)**: The cluster title and description contains information to answer part of the question.
 - **Distantly relevant (3)**: The cluster title and description does not contain information to answer the question, but the content relates vaguely to it.
 - **Less relevant (2)**: The cluster title and description does not contain information to answer the question, but some entities relate to it.
 - **Not relevant (1)**: The cluster title and description has no relation to the question.
 - Go to the provided **Result Sheet**. Answer the questions exactly as stated after each question on the HTML page.
 - If you need to verify your answers, Feel free to click on the **Source Code files** in the center workspace to open the specific project files directly on GitHub in a separate window
- or**
- A shorter version: > Click the subsystem link to open the relevant GitHub project section in a new tab or window. The **Source Code files**' names listed below indicate the files associated with the cluster and can be found in that GitHub location or its subdirectories.
- Record the evidence that helped you answer each question (*including the LLM-generated title, LLM-generated description, and any attached source code files*) and make every effort to find as much information as possible to answer the questions. *If you are not able to answer a question with the information presented to you, please state so clearly in the **Result Sheet**.*
6. Track your progress via the progress section in the left sidebar. There is one progress row for each task. Each dot represents one cluster. A dot turns **green** only when all three relevancy questions for that cluster and task have been rated. You must make all dots green for all tasks before downloading your results.
 7. The **Download (.JSON)** button is located at the bottom left of the interface. The button is disabled until all evaluations are complete. It becomes active only when all progress dots have been green.

8. Once the **Download (.JSON)** is activated, click it and a pop-up box will appear, in which you must enter your provided **Participant ID**. The download will not start until you:
 - Enter your Participant ID (e.g., G10M7)
 - Click the **Download** button in the pop-up box.
9. Please keep the downloaded .json file. (it will automatically be named based on The project name, the algorithm name, and your Participant ID (e.g., TIKa_ARC_G10M7_Results.json)).

Important note: Try to memorize if the project was easier or harder to answer its questions compared to other projects, as this will be asked in the *exit questionnaire*.

Vanilla LLM instructions

To ask an LLM, you will use a customized command-line tool, which we explain below. Before running the tool, please read the task and questions entirely to get an overview. After this, you can run the tool by following these steps:

1. Make sure you have the newest Java Runtime Environment installed. Run the tool using the following command in the command line:
`java -jar LLMConsole.jar`
2. You will be asked to provide the task ID and Question type (Existence, Property, and Executive). You can find the task ID in the title of each task, and the Question Type over each question. Please enter them correctly and press Enter.
3. You will then be allowed to ask a question to LLM. At that point, think about useful questions that can help you solve the task. Enter your question and press Enter once.
4. The LLM will produce an answer for you. Wait until the full answer has been generated.
5. You will be asked to determine the answer's relevance to the question. You need to select one evaluation from the following:
 - Very High Relevance (5): The whole answer contains useful and concise information that contains sufficient information to answer the entire question.
 - High Relevance (4): The whole answer contains useful and concise information that contains information to answer part of the question.
 - Medium Relevance (3): The answer contains different information. Some pieces of information can be useful to answer the question. Nevertheless, part of the answer might be redundant or irrelevant.

- Low Relevance (2): The answer contains different information. Few pieces of information are relevant to answering the question but might help to ask the next question. A large part of the answer might be redundant or irrelevant.
 - No Relevance (1): The whole answer has nothing to do with the question. It contains no relevant information.
6. After specifying the relevance, you will be asked to justify your evaluation. Explain in 1–2 sentences why you selected this rating.
 7. The tool saves all your answers in a file called data.csv, which is stored in the same folder as the JAR file.
 8. Repeat Steps 2–7 to ask more questions for either the same task or different tasks. Please always specify the correct task ID at the beginning.
 9. If you have finished asking the LLM, you can exit the tool by pressing Ctrl+C. You can restart the tool at any time by repeating Step 1. The data.csv file will be appended, meaning that previous answers will be preserved.
 10. Please send us the output data.csv file.

Important note: Try to memorize if the project was easier or harder to answer its questions compared to other projects, as this will be asked in the *exit questionnaire*.

After solving all tasks, please *fill in the exit questionnaire*, through this link:

Expected delivery of this assignment:

1. The **queries and evaluations** for each approach. This involves:
 - a) For the search engine, at least 2 queries per question need to be submitted and evaluated. Please submit the ratings .txt files on the PANDA course directly. We expect at least 36 files (one per query). Please name them to easily identify which project, task, and question in this format: Project_TaskName_Question_ID.
 - b) For the knowledge graph visualization, you will receive a .txt data file at the end of the tasks. This file should be submitted on the PANDA course directly.
 - c) For the vanilla LLM, the tool stores a .csv file with all information. This file should be submitted on the PANDA course directly.
 - d) For the clustered source code, the tool downloads a .json file containing your qualitative Likert-scale ratings for every cluster. This file should be submitted on the PANDA course directly.
2. The **Result Sheet** with information from issues, and evidence from the issues, such as text from issue description or comments, or attached documents or entities in the knowledge graph or answers of LLMs.
3. Filled **exit survey**, these are submitted automatically.

Evaluation

The assignment will be evaluated based on the following criteria:

1. The **correctness** and **completeness** of the delivery, especially making all tasks, questions, and using the correct approaches assigned to you.
2. The **reasonability** of the submitted queries, questions, and evaluated issues.
3. The **sufficiency** of the information in the **Result Sheet**.